

M S SAUJANYA

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OBJECTIVE

Passionate Electronics and Communication Engineering student with a strong foundation in VLSI and RTL design. Aspiring VLSI Engineer equipped with hands-on experience in Verilog, SystemVerilog, and digital design optimization, eager to contribute to cutting-edge semiconductor innovations. Driven problem-solver aiming to apply theoretical knowledge and simulation expertise to tackle complex hardware challenges.

TECHNICAL SKILLS

HDL / Languages	Verilog, SystemVerilog (basic), C/C++, Python
Verification	Gate-Level Simulation (GLS), Functional Simulation, Testbench Design
EDA Tools	ModelSim, Quartus Prime, Cadence Virtuoso, Yosys, VS Code
PDK / Technology	Sky130 (open-source 130nm PDK), Cyclone IV FPGA
Design Concepts	RTL Design, FSM, Pipelining, Low-Power Design
Protocols	SPI, AMBA, I2C, UART, Wishbone

PROJECTS

Synthesizable SPI Master Core in Verilog RTL Design — Maven Silicon

- Designed a fully synthesizable SPI Master Core in Verilog with WISHBONE-compliant host interface supporting SPI Mode 0–3 and Microwire/Plus protocols.
- Implemented clock polarity/phase control, shift register logic, and chip-select management in RTL.
- Verified functionality through ModelSim simulation and confirmed synthesis compatibility using Yosys.

Binary BCD Converters using CMOS and Pseudo-NMOS Logic Cadence Virtuoso

- Analyzed CMOS vs Pseudo-NMOS logic for 4-bit Binary-to-BCD converters using Cadence Virtuoso.
- Reduced power consumption by $5.2\times$ with CMOS design ($776.6\text{ }\mu\text{W}$) compared to Pseudo-NMOS (4.083 mW).
- Verified that power reduction maintained performance with nearly identical propagation delay ($\sim 48.66\text{ ps}$).

Iterative Computing Algorithm in FPGA using Altera CORDIC FPGA Implementation

- Implemented an iterative CORDIC algorithm on FPGA to compute sine and cosine functions.
- Evaluated performance across 8-bit, 12-bit and 16-bit fixed-point precision architectures.
- Demonstrated 8-bit design reached $\sim 110\text{ MHz}$ frequency, 41% faster than 16-bit ($\sim 78\text{ MHz}$).
- Analyzed hardware utilization: precision doubling increased ALUT usage ($75 \rightarrow 154$) and registers ($60 \rightarrow 130$).

Neural Network Accelerator on EEG Data for Seizure Detection Verilog — FPGA CNN Accelerator

- Designed RTL for a pipelined CNN accelerator in Verilog targeting Cyclone IV FPGA, using MAC units and FSM-based control logic for seizure detection from EEG data.
- Achieved post-synthesis frequency of 85.4 MHz with 92.3 mW power at 72% logic utilization.
- Validated RTL against a Python reference model; achieved 86.6% classification accuracy with $\sim 800\times$ speedup over software baseline.
- Wrote functional testbenches in Verilog to verify datapath and control logic independently before integration.

CERTIFICATIONS

SPI Design Maven Silicon

- Designed and implemented a fully synthesizable SPI Master Core in Verilog, featuring a WISHBONE-compliant host interface and compatibility with SPI and Microwire/Plus protocols.

Master RTL Design And Synthesis

VSD

- Designed synthesis-friendly Verilog RTL for combinational and sequential circuits; mapped to Sky130 standard cell library using Yosys.
- Performed Gate-Level Simulation (GLS) to detect and resolve synthesis-simulation mismatches; analyzed timing using standard delay files (SDF).
- Gained exposure to static timing analysis (STA) concepts including setup/hold violations and critical path optimization.

EDUCATION

Vellore Institute of Technology

2022 - 2026

B.Tech in Electronics and Communications Engineering

CGPA: 8.3 / 10.0

Sri Chaitanya

2020 - 2022

Telangana State Board of Education

Score: 98%

DAV Public School

2019 - 2020

Central Board of Secondary Education

Score: 92.4%

RELEVANT COURSEWORK

Digital Logic Design, VLSI Design, Embedded Systems, Computer Architecture, Signals & Systems, Analog Electronics, Microprocessors